Model conversion

Model conversion

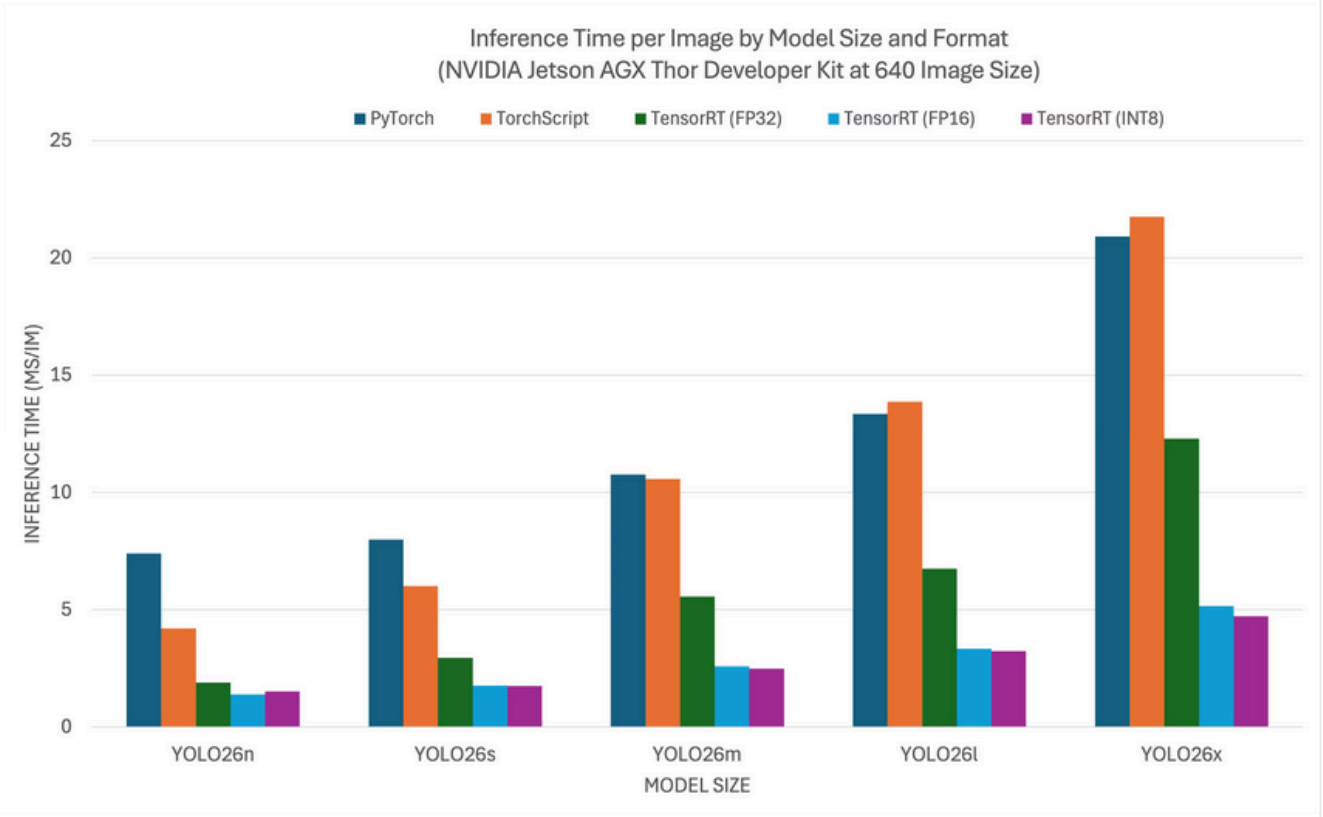
1. jetson yolo26 (the benchmark test)
2. Enable the best performance of the mainboard
   - 2.1. Enable the MAX power mode
   - 2.2. Enable the Jetson clocks
3. Model conversion
   - 3.1. CLI: pt → onnx → engine
   - 3.2. Python: pt → onnx → engine
4. Model prediction
   - The CLI use

References

1. jetson yolo26 (the benchmark test)

The yolo26 benchmark test data comes from the Ultralytics team, which tested models in many different formats (the data is for reference only).

NVIDIA Jetson AGX Thor Developer Kit



According to the test parameters of the different model formats provided by the Ultralytics team, we can see that the inference performance is the best when TensorRT is used!

Reference link:

```
https://docs.ultralytics.com/guides/nvidia-jetson
```

2. Enable the best performance of the mainboard

2.1. Enable the MAX power mode

Enabling the MAX Power Mode (the maximum power mode) on the Jetson ensures that all the CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable the Jetson clocks

Enabling the Jetson Clocks ensures that all the CPU and GPU cores run at the maximum frequency:

```
sudo jetson_clocks
```

3. Model conversion

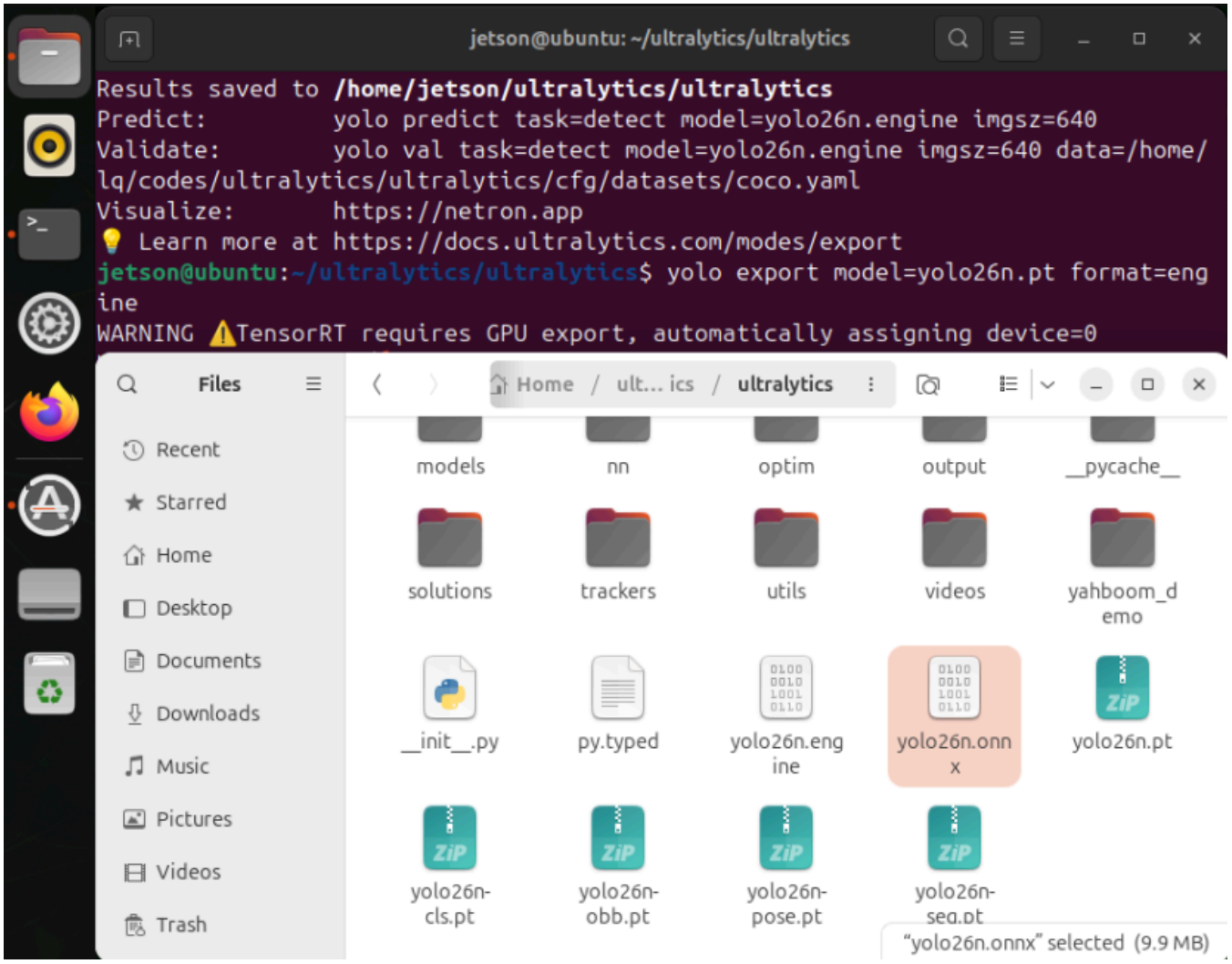
The first use of the export mode of yolo26 automatically installs some dependencies, just wait for the automatic completion! Choose either the CLI or the Python method.

3.1. CLI: pt → onnx → engine

Convert the model from the PyTorch format into TensorRT: the conversion process automatically generates the ONNX model

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo export model=yolo26n.pt format=engine
# yolo export model=yolo26n-seg.pt format=engine
# yolo export model=yolo26n-pose.pt format=engine
# yolo export model=yolo26n-cls.pt format=engine
# yolo export model=yolo26n-obb.pt format=engine
```



3.2. Python: pt → onnx → engine

Convert the model from the PyTorch format into TensorRT: the conversion process automatically generates the ONNX model

```
cd /home/jetson/ultralytics/ultralytics/yahboom_yolo26
```

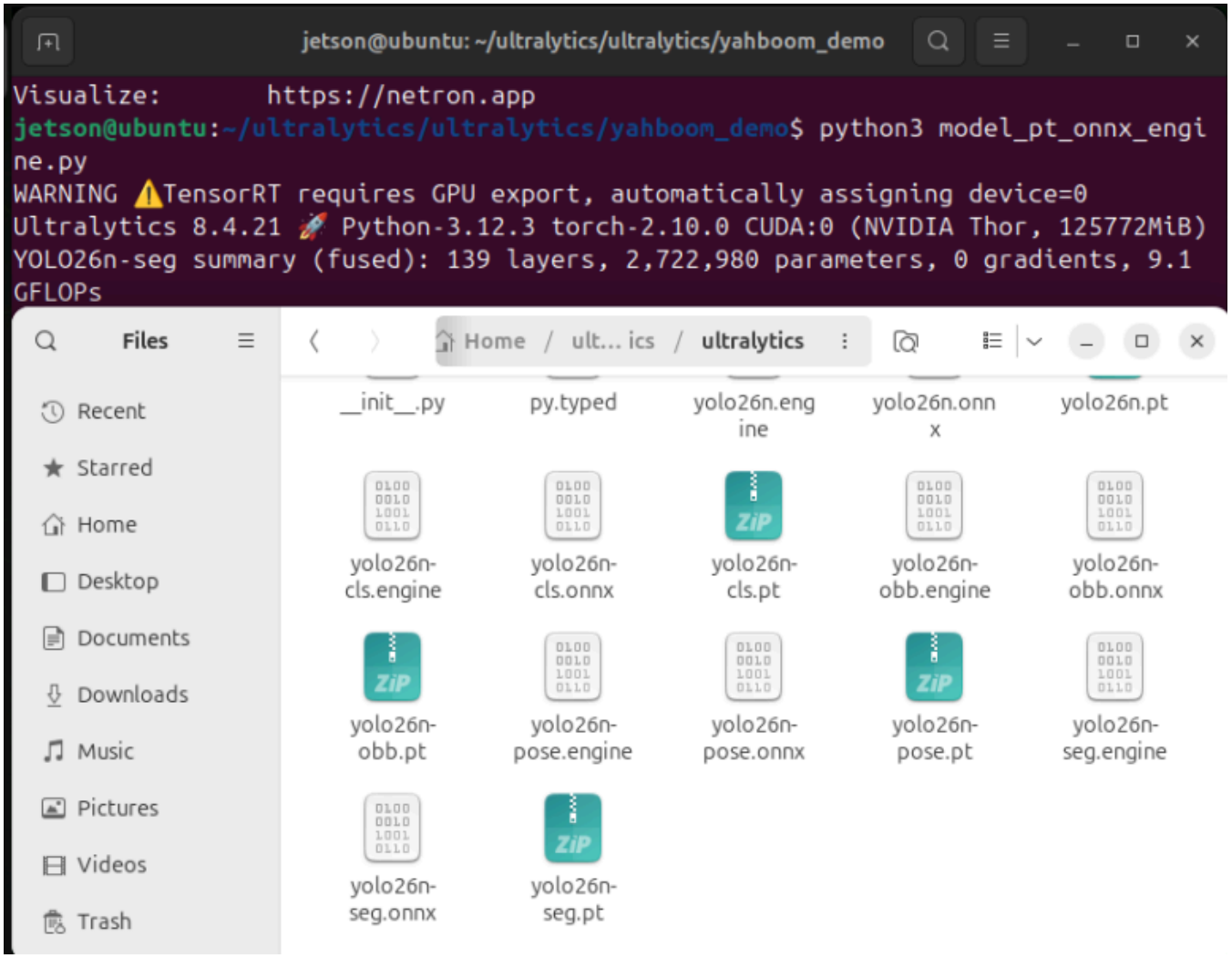
```
python3 model_pt_onnx_engine.py
```

```
from ultralytics import YOLO

# Load a yolo26n PyTorch model
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n.pt")
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-seg.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-pose.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-cls.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-obb.pt")

# Export the model to TensorRT
model.export(format="engine")
```

Note: the model file generated by the conversion is located at the position of the model file that was converted



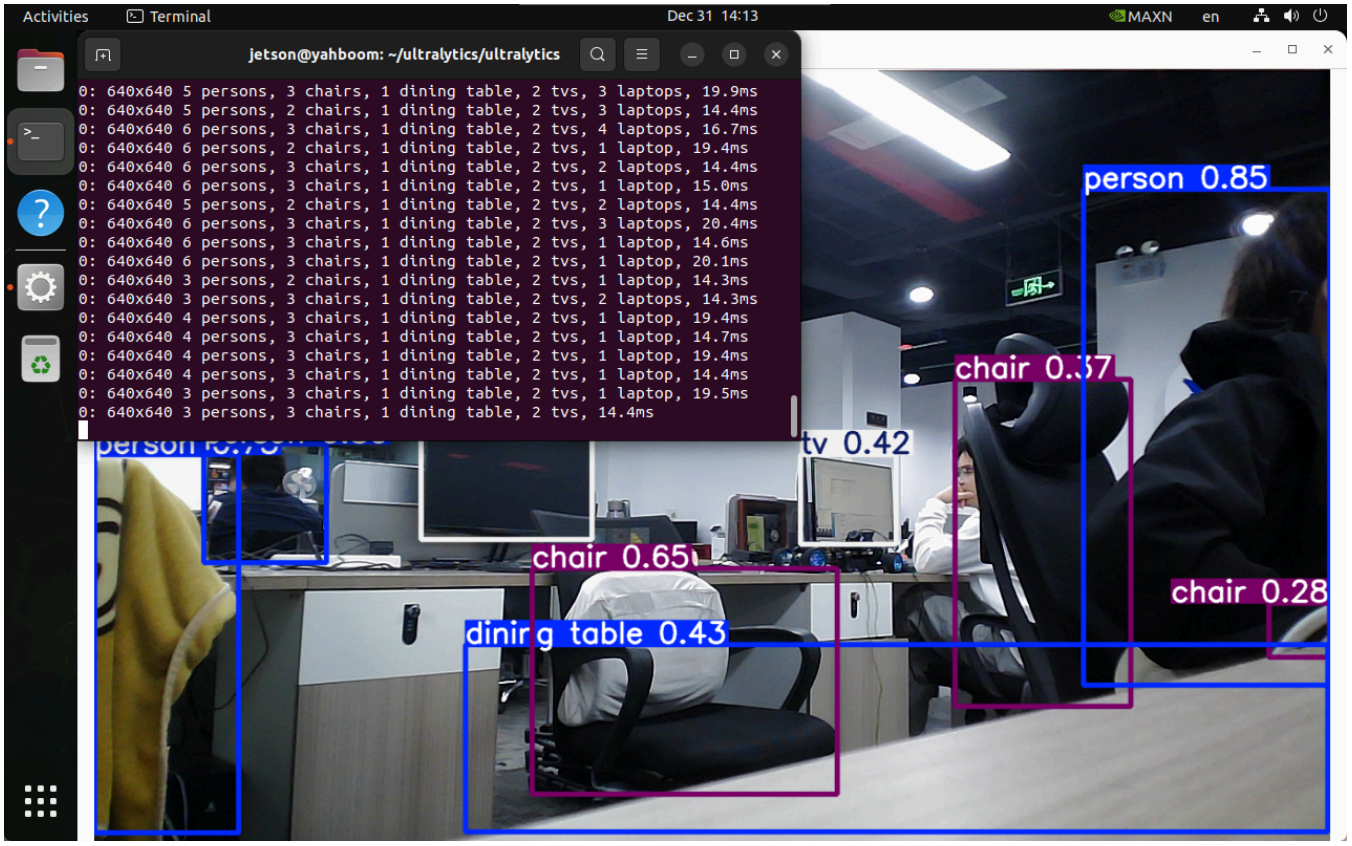
4. Model prediction

The CLI use

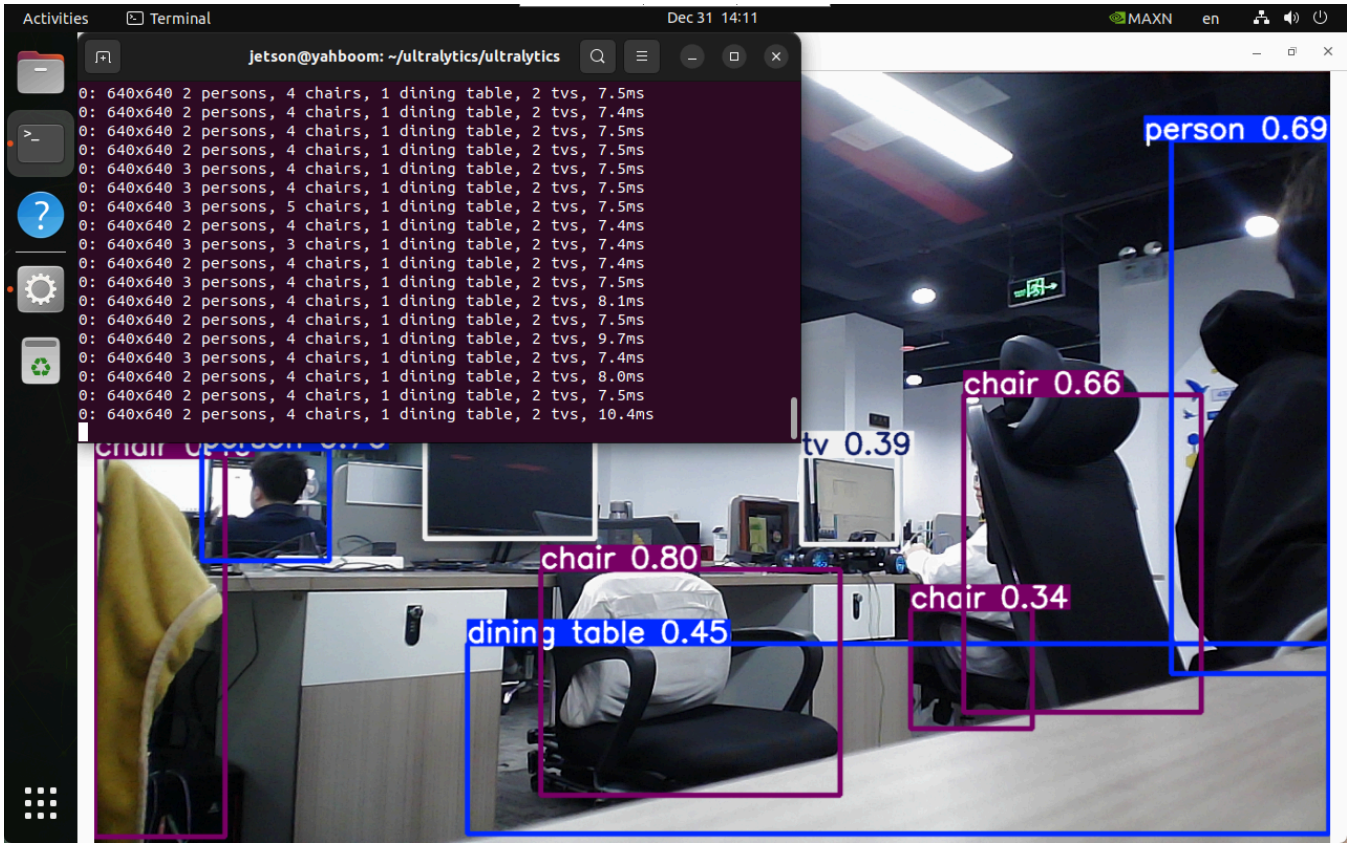
The CLI currently only supports calling the USB camera.

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo predict model=yolo26n.onnx source=0 save=False show
```




```
yolo predict model=yolo26n.engine source=0 save=False show
```



References

https://docs.ultralytics.com/guides/nvidia-jetson/

https://docs.ultralytics.com/integrations/tensorrt/